

CS & IT ENGINEERING



Computer Network

IPv4 Address

Lecture No. - 05



By - Abhishek Sir



Recap of Previous Lecture



Topic

Classful IPv4 Address

Topic

Subnetting





Topics to be Covered



Topic

Subnetting

(V. Imp.)



ABOUT ME



Hello, I'm **Abhishek**

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Topic : Subnetting



- Dividing (logically) a network into smaller manageable sub-networks
- Sub-network (subnet) : Clustering of hosts inside a network
- Clustering of hosts based on (some bits of host identifier) (HostId) field
[In practice, most significant bits of host identifier]



Topic : Subnetting



→ Before subnetting, (IP address) having two sections:

1. Network Identifier (Net ID) : x bits

2. Host Identifier (Host ID) : y bits

→ Size of IP address field = (x + y) bits



Topic : Subnetting



→ After subnetting, IP address having three sections :

1. Network Identifier (Net ID) : x bits
2. Sub-network Identifier (Subnet ID) : y₁ bits
3. Host Identifier (Host ID) : y₂ bits

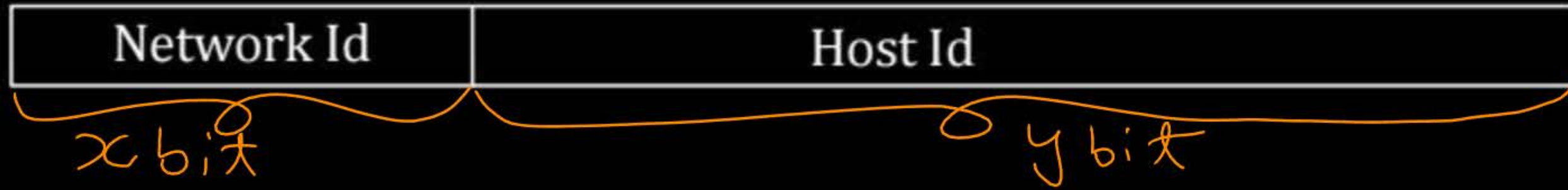
→ Size of IP address field = (x + y₁ + y₂) bits
[y = y₁ + y₂]



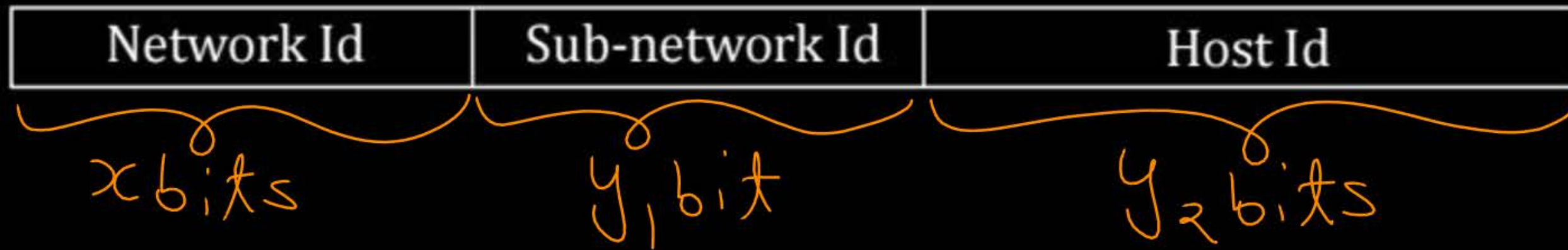
Topic : Subnetting



Before Subnetting



After Subnetting





Topic : Sub-network Address



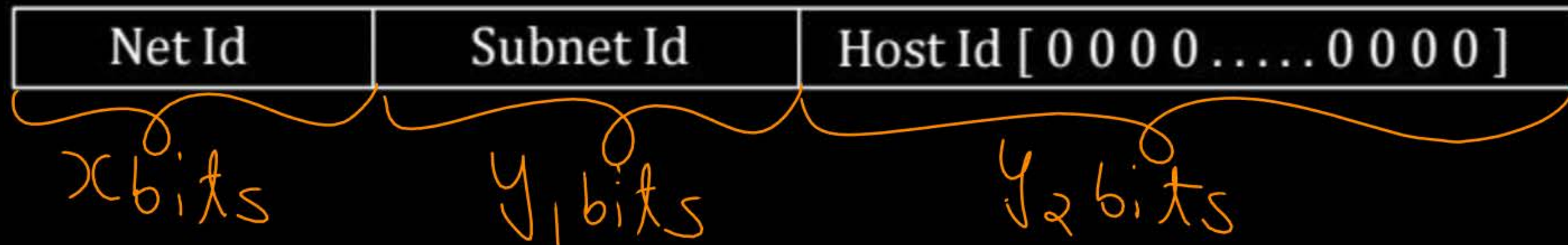
→ Special IP address (32 bits)

→ Used to represent a sub-network

NetID field = [As Assigned]

Subnet ID = [Anything]

HostID field = All Zero Bits





Topic : Sub-network Broadcast Address

→ Special IP address (32 bits)

→ Used to broadcast a packet to all hosts belongs to a sub-network

NetID field = [As Assigned]

Subnet ID = [Anything]

HostID field = All One Bits

Net Id	Subnet Id	Host Id [1 1 1 1 1 1 1 1]
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x bits

y_1 bits

y_2 bits



Topic : Host IP Address



→ Host IP address (32 bits)

→ Used to identify a host uniquely world wide

NetID field (x bits) = [As Assigned]

Subnet ID field (y_1 bits) = [Anything]

HostID field (y_2 bits) = [Any thing]
[Except all zero and all one bits]



Class C network (Network Address : 200 . 200 . 200 . 0), [with 3-bit subnetting.]
(8 bit Host ID)

Network Address : 200 . 200 . 200 . 0
Broadcast Address : 200 . 200 . 200 . 255
(Network Directed)
Network Mask : 255 . 255 . 255 . 0

[00000000]
[11111111]

Before Subnetting :-

First Host IP Address : 200 . 200 . 200 . 1
Last Host IP Address : 200 . 200 . 200 . 254

Network Size = $(2^8 - 2) = 254$ hosts

[00000001]
[11111110]

After 3-bit Subnetting :-

Sub-network Address

24 bit NetID 3-bit SubnetID 5 bit HostID

: 200 . 200 . 200 . _ _ _ 0 0 0 0 0

First Host IP Address

: 200 . 200 . 200 . _ _ _ 0 0 0 0 1

Last Host IP Address

: 200 . 200 . 200 . _ _ _ 1 1 1 1 0

Subnet Broadcast Address
(Sub-network Directed)

: 200 . 200 . 200 . _ _ _ 1 1 1 1 1

First Sub-network Address : 200 . 200 . 200 . 0
First Host IP Address : 200 . 200 . 200 . 1
Last Host IP Address : 200 . 200 . 200 . 30
First Subnet Broadcast Address : 200 . 200 . 200 . 31
(Sub-network Directed)

3-bit Subnet 5-bit Host ID → 
[000 00000]
[000 00001]
[000 1110]
[000 1111]

Second Sub-network Address : 200 . 200 . 200 . 32
First Host IP Address : 200 . 200 . 200 . 33
Last Host IP Address : 200 . 200 . 200 . 62
Second Subnet Broadcast Address : 200 . 200 . 200 . 63
(Sub-network Directed)

[001 00000]
[001 00001]
[001 1110]
[001 1111]

Last Sub-network Address : 200 . 200 . 200 . 224
 First Host IP Address : 200 . 200 . 200 . 225
 Last Host IP Address : 200 . 200 . 200 . 254
 Last Subnet Broadcast Address : 200 . 200 . 200 . 255
 (Sub-network Directed)

[111 00000]
 [111 00001]
 [111 1110]
 [111 1111]

Sub-network Mask : 255 . 255 . 255 . 11100000 = 255 . 255 . 255 . 224

Sub-network Size = $(2^5 - 2) = 30$ hosts per subnet

Network Size = $2^3 * (2^5 - 2) \text{ hosts} = 240 \text{ hosts}$



Last octet

1st subnet $\rightarrow$ 0 to 31

2nd subnet $\rightarrow$ 32 to 63

3rd subnet $\rightarrow$ 64 to 95

4th subnet $\rightarrow$ 96 to 127

5th subnet $\rightarrow$ 128 to 159

6th subnet $\rightarrow$ 160 to 191

7th subnet $\rightarrow$ 192 to 223

8th (last) subnet $\rightarrow$ 224 to 255



$$\begin{array}{r} 200 \cdot 200 \cdot 200 \cdot 50 \\ 255 \cdot 255 \cdot 255 \cdot 0 \\ \hline 200 \cdot 200 \cdot 200 \cdot 0 \end{array}$$

$$\begin{array}{r} 200 \cdot 200 \cdot 200 \cdot 50 \\ 255 \cdot 255 \cdot 255 \cdot 224 \\ \hline 200 \cdot 200 \cdot 200 \cdot 32 \end{array}$$

$$\begin{array}{r} 200 \cdot 200 \cdot 200 \cdot 60 \\ 255 \cdot 255 \cdot 255 \cdot 224 \\ \hline 200 \cdot 200 \cdot 200 \cdot 32 \end{array}$$

$$\begin{array}{r} 60 \rightarrow 00111100 \\ 224 \rightarrow 11100000 \\ \hline 32 \leftarrow 00100000 \end{array}$$

$$\begin{array}{r} 50 \rightarrow 00110010 \\ 224 \rightarrow 11100000 \\ \hline 32 \leftarrow 00100000 \end{array}$$



Topic : Sub-network Mask



→ Sub-network Mask (Subnet mask) [32-bits]

NetID field = [All One Bits]

Subnet ID field = [All One Bits]

HostID field = [All Zero Bits]

→ Used to generate subnet address from a given IP address



[MCQ] (H.W.)

IIT-M

[GATE-2019][2 Mark]



#Q. Consider three machines M, N and P with IP addresses "100 . 10 . 5 . 2", "100 . 10 . 5 . 5" and "100 . 10 . 5 . 6" respectively. The subnet mask is set to "255 . 255 . 255 . 252" for all the three machines. Which one of the following is true?

- ☐ A M, N and P all belong to the same subnet
- ☐ B Only N and P belong to the same subnet
- ☐ C Only M and N belong to the same subnet
- ☐ D M, N, and P belong to three different subnets



Topic : Size of subnet



→ Sub-network Size : Maximum number of hosts can be in a sub-network

For IP address (32 bits)

NetID field (x bits)

Subnet ID field (y₁ bits)

HostID field (y₂ bits) ←

Sub-network Size = [2^{y₂} - 2] hosts per subnet

[MCQ]

[GATE-IT-2004][1 Mark]



#Q. A subnet has been assigned a subnet mask of 255.255.255.192. What is the maximum number of hosts that can belong to this subnet?

- ☐ A 14
- ☐ B 30
- ☒ C 62
- ☐ D 126

Ans: C

Subnet Mask :

$$\underline{255.255.255.192} = 255.255.255.\underbrace{11000000}_{6 \text{ bit Host ID}}$$

Number of bits for HostID field = (y₂ bits) = 6 bits

Maximum number of hosts can be in this Sub-network

$$= [2^{y_2} - 2] \text{ hosts per subnet}$$

$$= [2^6 - 2] \text{ hosts}$$

$$= 62 \text{ hosts}$$

[MCQ]

[GATE-CS-2008][2 Mark]



#Q. If a class B network on the Internet has a subnet mask of 255 . 255 . 248 . 0, what is the maximum number of hosts per subnet?

☐ A 1022

☐ B 1023

☒ C 2046

☐ D 2047

Ans: C

Subnet Mask :

$$\underline{255.255.248.0} = 255.255.11111\underbrace{000000000000}_{11 \text{ bit HostID}}$$

Number of bits for HostID field = (y_2 bits) = 11 bits

Maximum number of hosts can be in this Sub-network

$$= [2^{y_2} - 2] \text{ hosts per subnet}$$

$$= (2^{11} - 2) \text{ hosts}$$

$$= 2046 \text{ hosts}$$



Topic : Number of subnets



For IP address (32-bits)

NetID field (x bits)

Subnet ID field (y₁ bits) ←

HostID field (y₂ bits)

Number of subnets = $[2^{y_1}]$ subnets per network

[* Subnet Id all zero and all one allowed]

[MCQ]

[GATE-CS-2005][1 Mark]



#Q. An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be :

- A 255 . 255 . 0 . 0
- B 255 . 255 . 64 . 0
- C 255 . 255 . 128 . 0
- ✓ D 255 . 255 . 252 . 0

Ans: D

Class B Network :-

Number of bits for Network ID field = [16 bits]

Number of Subnets = [64]
[For 64 departments]

Number of bits for Subnet ID field = $\lceil \log_2(\text{Number of Subnets}) \rceil \text{ bits}$

Subnet Mask = ~~255.255.~~ = $\lceil \log_2(64) \rceil \text{ bits} = 6 \text{ bits}$

= $\underbrace{255.255.}_{16 \text{ bit NetID}} \underbrace{11111100}_{6 \text{ bit subnetID}} \cdot \underbrace{00000000}_{10 \text{ bit hostID}}$
= 255.255.252.0



Topic : Size of network



→ Network Size : Maximum possible number of hosts can be in a network

For Host IP address (32-bits)

NetID field (x bits)

Subnet ID field (y_1 bits) ←

HostID field (y_2 bits) ←

Network Size = Number of Subnets per network * Sub-network size

= $[2^{y_1}]$ * $[2^{y_2} - 2]$ hosts per network



Topic : Reduction in networks size



Network Size (before subnetting)

$$\boxed{S_1} = [2^{(y_1+y_2)} - 2] \text{ hosts per network}$$

Network Size (after subnetting)

$$\begin{aligned}\boxed{S_2} &= [2^{y_1}] * [2^{y_2} - 2] \text{ hosts per network} \\ &= [2^{(y_1+y_2)} - 2^{(y_1+1)}] \text{ hosts per network}\end{aligned}$$

Reduction in Network Size (after subnetting)

$$\boxed{S_1} - \boxed{S_2} = [2^{(y_1+1)} - 2] \text{ hosts per network}$$

$$= (2 * 2^{y_1} - 2) \text{ hosts}$$

[MCQ] H.W.

[GATE-IT-2006] [2 Mark]



#Q. A subnetted Class B network has the following broadcast address : 144 . 16 . 95 . 255. Its subnet mask

- ☐ A is necessarily 255 . 255 . 224 . 0
- ☐ B is necessarily 255 . 255 . 240 . 0
- ☐ C is necessarily 255 . 255 . 248 . 0
- ☐ D could be any one of 255 . 255 . 224 . 0, 255 . 255 . 240 . 0, 255 . 255 . 248 . 0



2 mins Summary



Topic

Subnetting



THANK - YOU

